Working from the Command Line on Quest

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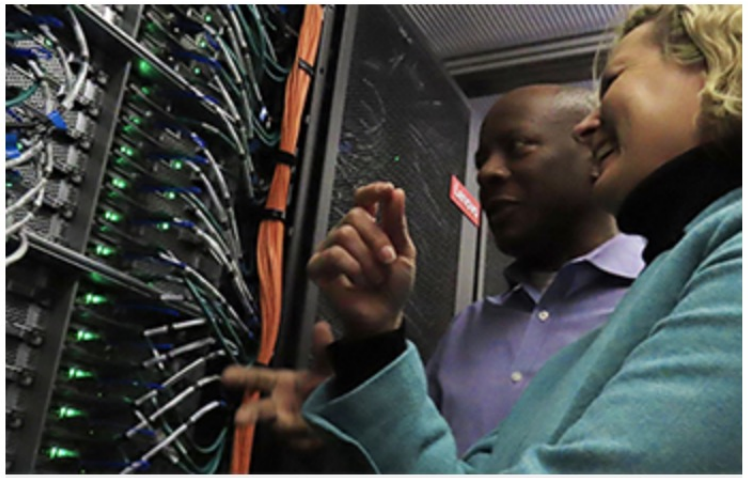

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Access to high performance computing, research software, and global networks for conducting computationally intense research.

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Support for collecting, analyzing, visualizing, and programming with research data.

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Topics

- Introduction to the Quest system
- Working from the command line - filesystem navigation
- Working from the command line - file editing
- Submitting jobs through the scheduling system

Materials:

https://github.com/nuitrcs/gcc_command_line

Quest: Northwestern’s HPC cluster

- HPC = high performance compute
- All researchers at Northwestern can use General Access Quest resources.
- Why would you use it?
  - more computational power than your laptop
  - more storage than your laptop
  - shared workspace for collaboration
  - Linux based environment, required by some research software

Log on to Quest

```
ssh <netid>@login.quest.northwestern.edu
```

- if this is the first time, type yes after being prompted to fingerprint
- type your NetID password (it doesn’t look like you’re typing, but you are) and hit “Enter”

Navigating the filesystem: pwd

Type pwd and hit “Enter”.

Should return: /home/<netid>

- pwd stands for print working directory
- It displays the path to your current location in the filesystem.

Home directory: /home/<netid>

- The “root” folder of a Linux (or Mac) system is /
- All subsequent folders are located under /
- You are currently within your **home directory**, which is a directory with your NetID, under the home directory.
- Exclusively your space, 80Gb of storage
- Only portion of Quest’s filesystem that is backed up
- Recommended location for personal software installs, code, and job scripts.

Navigating the filesystem: `cd`

```
cd /projects/e30682/Quest_Intro
```

- `cd` stands for change directory (folder)
- Used to move from one location to another in the filesystem
- A couple of shortcuts: `cd` without a location and `cd ~` will both take you to your home directory, `cd ..` will take you one directory towards root, or “up” one directory

Navigating the filesystem: `ls`

```
ls
```

- `ls` stands for list
- Used to list the contents of a directory
- Without any other parameters, it will list the contents of your current working directory. But you can point it to other locations.

Navigating the filesystem: `ls`

```
ls -al
```

- The -a flag means “all” and will include hidden files.
- The -l flag means “long form” and will print additional information in a long form list, instead of just file and folder names.
- There are many other possible parameters, the command man followed by the command you’re interested in learning more about will pull up a manual for you.

A little practice…

- Make sure you're in `/projects/e30682/Quest_Intro`
- What is the content of this directory?
- What is the content of the directory `explore`?
- Run `ls ./explore/labyrinth/stable/donkeys`
  - Run the same command as above, while passing the `-a` and `-l` flags together.
  - What is different about the output?
- How could you get to `/projects/e30682/Quest_Intro/explore/labyrinth`?

A little practice...

- How could you get to `/projects/e30682/Quest_Intro/explore/labyrinth`?
- Run `cd explore/labyrinth`
- Run `cd ..` Where are you now?
- Go to `explore/labyrinth/garden_path`. What are the folders?
- Go into the `bugs` folder. Now, how could you get to the `rocks` folder?

Examining files: `cat`, `head`, `tail`

`cat <filename>`

- Stands for concatenate
- Prints the contents of a file to the console
- `head` and `tail` will print the beginning and the end, respectively, not the whole file
- Only helpful for human-readable files (eg. .txt, .md, .R, .py) not things like .tar, .gz, .zip

Explore the labyrinth...

- Answer the following questions and look for secret tips.
- Find the coins in the fountain in the labyrinth. What coins are in there? What is the full path to the file `nickles.md`?
- What is in the fridge in the kitchen in the hobbit house in the meadow in the wooded lane in the labyrinth?
- What are the names of our ponies? What is the full path to where the ponies live?
- How would you get from the ponies to the horses?
- Find the hidden treasure.

Questions?

quest-help@northwestern.edu

Storage folders on Quest

- /home/<netid> - your home directory, 80GB of space, individual use
- /scratch/<netid> - individual scratch storage space, 5TB, available by request
- /projects - project-based storage folders on Quest
  - /projects/e30682 - shared persistent storage for this course
  - /projects/b1042 - shared scratch storage for the Genomics Compute Cluster
  - /projects/genomicsshare - shared read-only storage for genomics reference files
  - /projects/pXXXXX - per project persistent storage space of 1TB, available by request
- scratch storage spaces always has an automatic deletion process
- home is the only space on Quest that is backed up

File editing from the command line: `mkdir`

```
mkdir <directoryname>
```

- Creates a new directory with the name provided to the command

Try it! Within `/projects/e30682/Quest_Intro`, make your own folder.

File editing from the command line: nano

```
nano <filename>
```

- Opens the file in a command-line editor
- Allows you to write or edit files from within the terminal
- Ctrl+O, Ctrl+X to save and exit

Try it! From within your new folder, create a new file with nano called `notes.txt`.

File editing from the command line: nano

```
nano notes.txt
```

Write some notes about the commands we've used today.

Use *Ctrl+O*, *Ctrl+X* to save and exit.

File editing from the command line: `cp`

```
cp </path/to/file/filename> <desired/location/>
```

- Copies a file from an existing location to a new location
- You can rename files if you provide a new name with your new location.
- You can copy an entire folder if you add `-r` or `-R`

Try it! From within your new folder,

```
cp /projects/e30682/Quest_Intro/files/introduction.txt
```

to your new folder.

File editing from the command line: `mv`

```
mv <previousfilename> <newfilename>
```

- Stands for “move”.
- Can be used to move a file or rename a file.
- It is always safer to `cp` and then `rm` the old file than to `mv` it directly, but this is more of a concern for larger files than small pieces of text.

Try it! Rename your new copy of `introductions.txt` to `<yourname>.txt`.

A little practice...

- Open your new copy of `introduction.txt` with *nano*.
- Edit the file with your name and favorite color.
- Save and exit from nano.
- Copy the newly edited file into `/projects/e30682/Quest_Intro/`.

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MacBook Pro M3

compute:
36 GB RAM
16 cores

storage:
500 G disk space

Users limited to:

compute:
6 GB RAM
4 cores

storage:
80 G disk space /home
1024+ G disk space /projects

Per node max:

compute:
1970 GB RAM
128 cores

storage:
80 G disk space /home
1024+ G disk space /projects

Submitting jobs: `srun`

```
srun -A e30682 -p short -t 00:10:00 --pty bash -l
```

- Requests an interactive job from the scheduling software (Slurm)
- Requires the following parameters
  - `-A/--account` the account/group you're making the reservation under
  - `-p/--partition` the group of computers (nodes) you're submitting to
  - `-t/--time` the amount of time allotted for the job
  - `--pty bash -l` setting up a log in type session, must be at the end

Try it! Use the command above the bullet points.

Submitting jobs: `srun`

```
srun -N 1 -n 4 --mem=100G -A e30682 -p short -t
00:10:00 --pty bash -l
```

- Additional parameters allow you to take advantage of the cluster resources:
  - `-N/--nodes` the number of nodes you're reserving
  - `-n/--ntasks` the number of cores (CPUs) you're reserving
  - `--mem` the amount of memory allotted for the job (default 3G per core)

Partitions and resources

Partition	Time Limit	Core Limit	RAM Limit	Description
short	4 hours	128	450G	CPU resources
normal	48 hours	128	450G	CPU resources
long	7 days	128	450G	CPU resources
genhimem	7 days	52	1400G	CPU resources

*GPU resources are available in the gengpu queue.
More information about partitions:
https://rcdsdocs.it.northwestern.edu/systems/quest/user-guide/slurm/slurm.html#partitions

Submitting jobs: `sbatch`

```
sbatch <jobscript.sh>
```

- Schedules a batch job through the scheduling software (Slurm).
- Uses the same parameters as `srun` (except `--pty bash -l`) but uses a *script* instead of a command.
- Allows you to automate tasks instead of running them interactively.
- Allows you to scale in a more high-throughput manner.

What does a jobscript look like?

```
#!/bin/bash
#SBATCH -A e30682
#SBATCH -p short
#SBATCH --time=00:10:00
#SBATCH -N 1
#SBATCH -n 1
#SBATCH --mem=5G
```

Parameters for scheduling with Slurm

```
module purge
module load fastqc/0.12.0

fastqc sample_01.fastq.gz
```

Everything you would have typed interactively.
In this case, loading a software module for a quality control program and then running that program.

Try it!

1. Copy `/projects/e30682/Quest_Intro/files/example_script.sh` to your folder.
2. Open the file with `nano`.
3. Edit the SBATCH parameters as follows:

   ```
   account=e30682
   partition=short
   time=00:15:00
   ```

4. Save and exit nano.
5. Launch the edited job script from your folder:

   ```
   sbatch example_script.sh
   ```

Submitting jobs: squeue

```
squeue -u <netid>
```

- Shows the jobs in the queue for the given user.
- Queued jobs have the status Pending, or Running.
- Pending jobs will list the reason they are waiting
  - Resources means you are next in line but the resources aren’t available
  - Priority means a different job in the queue has higher priority

Submitting jobs: `sacct`

```
squeue -X -S <startdate>
```

- Shows your job history.
- Default to only today's history if no start date is provided.
- Start dates should be provided in MMDDYY.
- This is a good way to look up Job ID numbers.

Submitting jobs: `seff` / `checkjob`

```
seff <jobid>
```

- Shows a job's resource use efficiency
- Only accurate after a job has completed

```
checkjob <jobid>
```

- Shows job details including, seff output
- Also includes Slurm variables and the job script

More practice...

1. Copy `/projects/e30682/Quest_Intro/files/bwa.sh` to your folder.
2. Open the file with `nano`.
3. Edit the SBATCH parameters
    1. Use our classroom account
    2. Choose resources based on `seff` output
4. Save and exit nano.
5. Launch the edited job script from your folder:

    ```
    sbatch bwa.sh
    ```

```
(base) [hsc945@quser33 haley]$ seff 4852999
Job ID: 4852999
Cluster: quest
User/Group: hsc945/hsc945
State: COMPLETED (exit code 0)
Nodes: 1
Cores per node: 8
CPU Utilized: 03:46:25
CPU Efficiency: 97.87% of 03:51:20 core-walltime
Job Wall-clock time: 00:28:55
Memory Utilized: 6.11 GB
Memory Efficiency: 40.73% of 15.00 GB (15.00 GB/node)
```

Questions?

quest-help@northwestern.edu

Software on Quest: `module`

```
module spider <software>
```

- Searches our module system by software name
- Returns modules matching the name provided, and the versions available

```
module load <software>/<version>
```

- Loads a module so that that software is available in your environment
- It’s always best practice to specify a version

Software on Quest: `module`

```
module list <software>
```

- Lists modules currently loaded
- Without a software name, returns all modules loaded
- With a software name, returns only those matching that name

Software on Quest: `module`

```
module unload <software>/<version>
```

- Unloads a specific module

```
module purge
```

- Unloads all modules

Software on Quest: `which`

```
which <command>
```

- Tells you where a command you want to use is coming from
- Could point to a module
- Could point to a “system-level” install
- Could point to a virtual environment
- Will return `no <command> in $PATH` if that command is not in your environment

Try it!

1. Check to see which version of R, or if a version of R is available in your environment.
2. Search for the versions of R we have installed as modules.
3. Load the newest available version of R.
4. Check what other modules this loads.
5. Does this load a version of jpeg?

Questions?

quest-help@northwestern.edu

How can you interact with Quest after this course?

- Ask your PI if they have an existing Quest Allocation you should join
- Request a new General Access Allocation for a specific research project
- Apply to join the Genomics Compute Cluster
- Email quest-help@northwestern.edu with any questions!
- Reach out to haley.carter@northwestern.edu